

Auteur: Stan Van Campenhout

Studierichting: Informatica

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$$\int \frac{1}{x} (\ln(x))^n dx = \int (\ln(x))^n d(\ln(x))$$

$$\stackrel{n \neq -1}{=} \frac{(\ln(x))^{n+1}}{n+1} + c$$

$$\stackrel{n = -1}{=} \ln |\ln(x)| + c$$

$$\text{Als } n \neq -1 \Rightarrow \int \frac{1}{x} (\ln(x))^n dx = \frac{(\ln(x))^{n+1}}{n+1} + c$$

$$\text{Als } n = -1 \Rightarrow \int \frac{1}{x} (\ln(x))^n dx = \ln |\ln(x)| + c$$

References